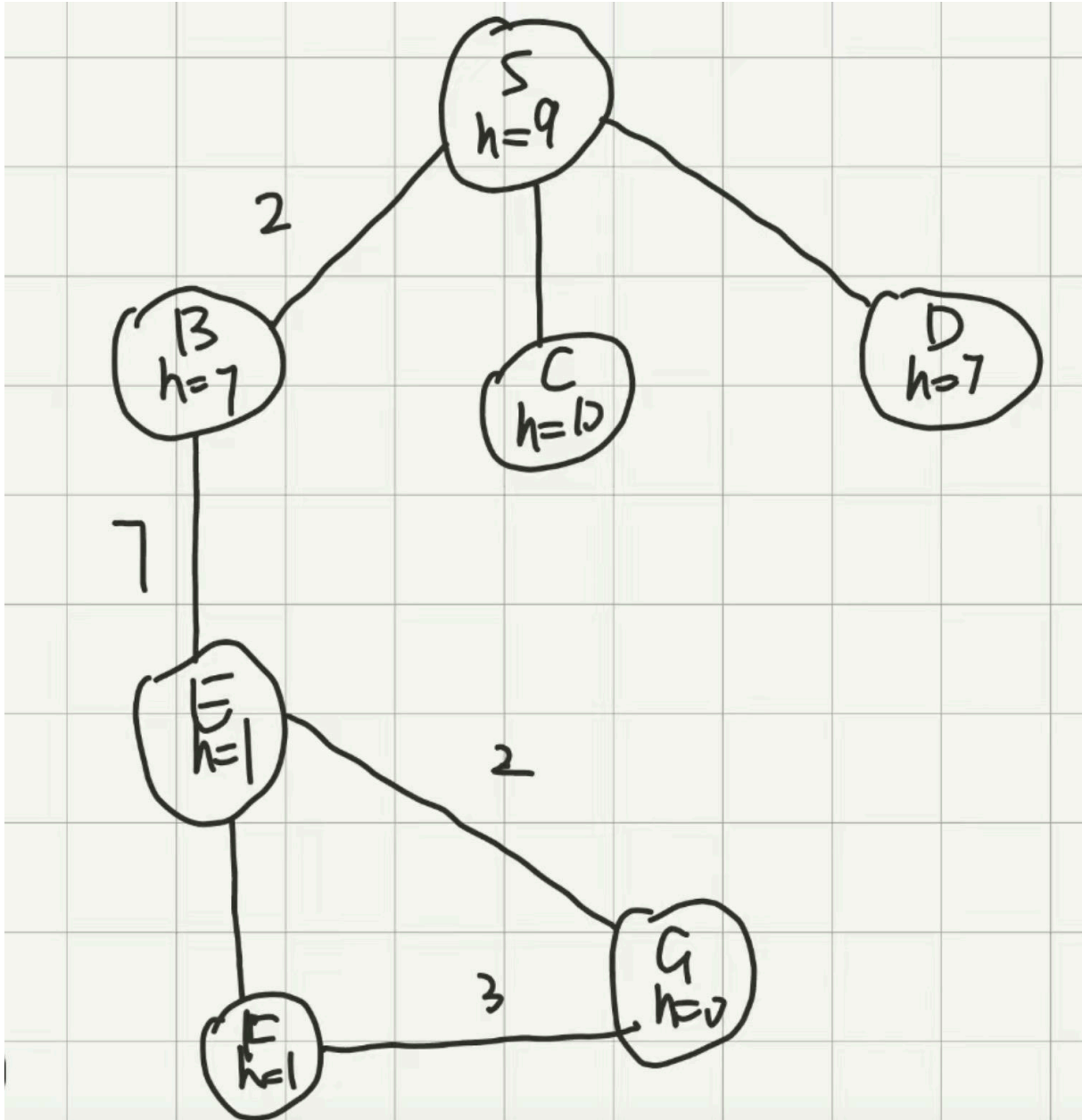


Artificial Intelligence 2024Spring – Final Exam

SuperbRain

2024-06-26

Problem 1: Search

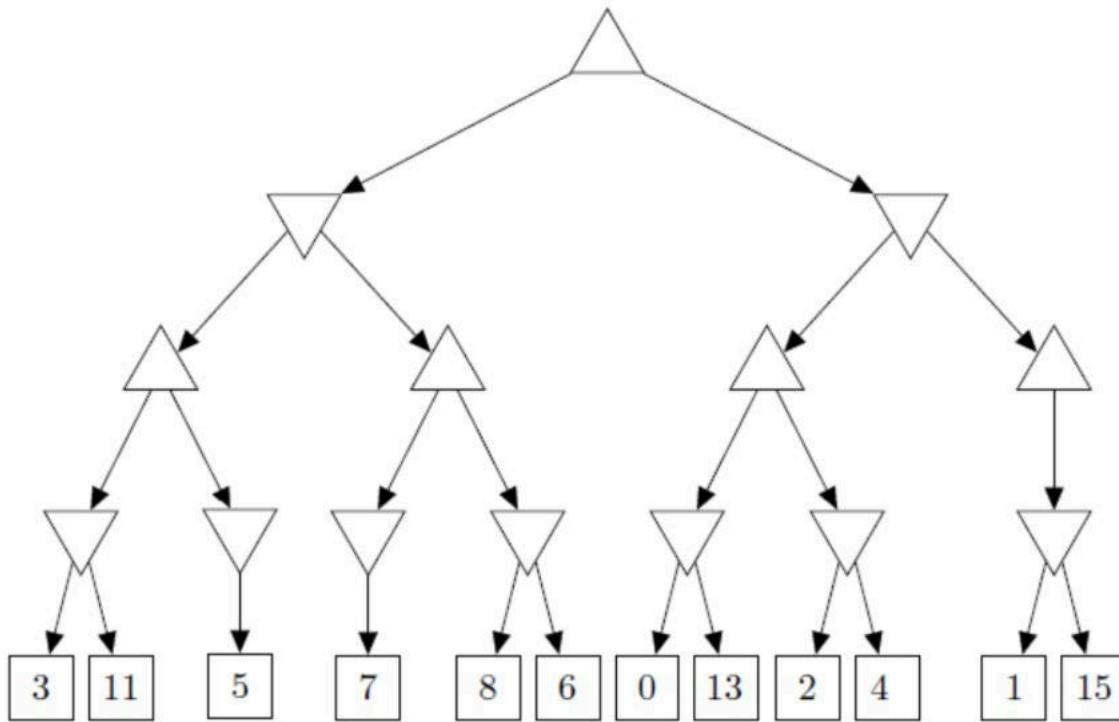


(1) Uniform-cost search

(2) Best first search

(3) A^* search

Problem 2: Minimax Search



(Numbers change to 11, 3, 5, 7, 8, 15, 0, 13, 2, 4, 1, 6)

alpha-beta pruning

Problem 3: CSP

$$X = \{X_1, X_2, X_3\}$$

$$D(X_1) = \{1, 2, 3\}, D(X_2) = \{1, 2, 3\}, D(X_3) = \{2, 3\}$$

$$C_1 : X_1 \neq X_2, C_2 : X_2 < X_3, C_3 : X_1 \neq X_3$$

- (1) Draw the constraint graph for the given CSP;
- (2) $X_1 = 2$, forward checking;
- (3) $X_1 = 2$, arc consistency;
- (4) Given a solution for this CSP or states that none exists.

Problem 4: Knowledge and Reasoning

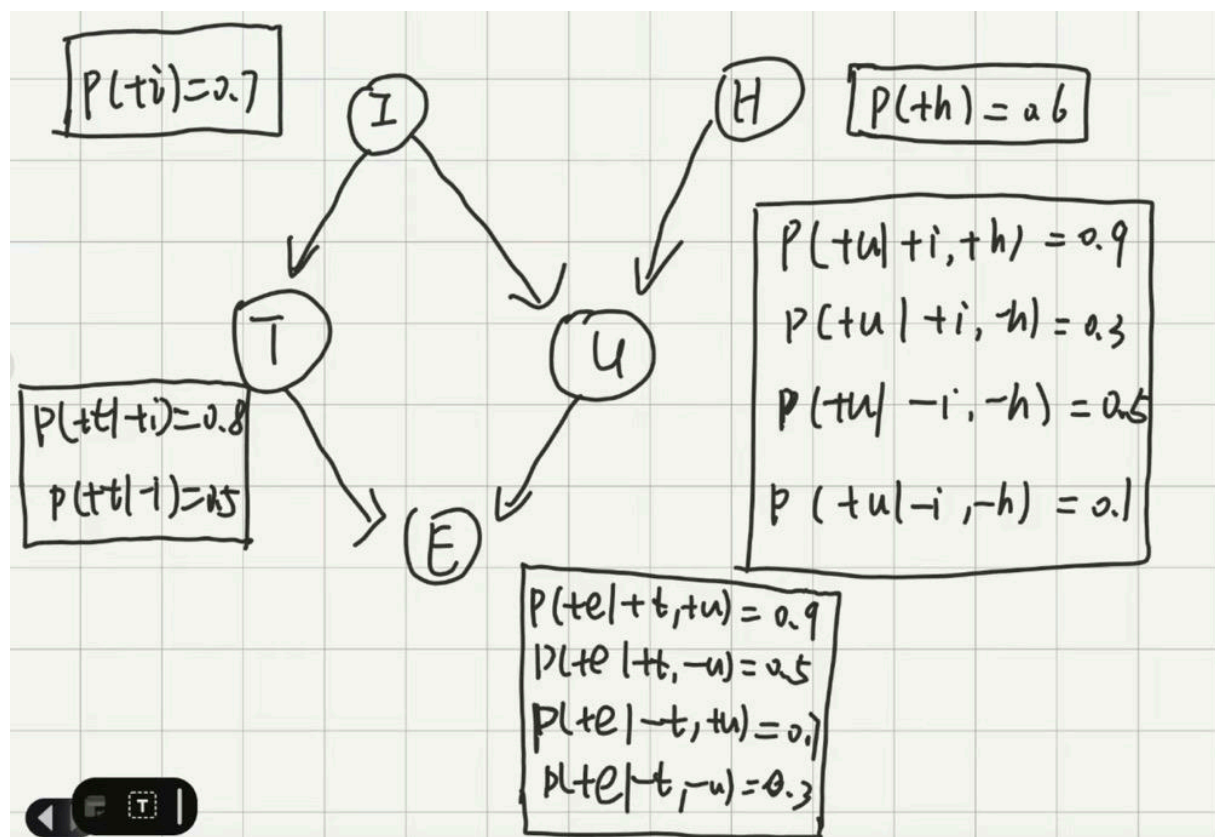
$A \wedge \neg I, W \Rightarrow I, \neg A, \neg W \wedge \neg I \Rightarrow F$, Prove F .

Problem 5: Decision Tree

Age	Temperature	Headache	Diagnosis
20-30	High	Yes	Positive
40-50	High	No	Positive
30-40	Normal	No	Negative
40-50	Low	No	Negative
20-30	Normal	Yes	Positive
50-60	High	No	Positive
40-50	Normal	No	Negative
20-40	Low	Yes	Negative

- (1) Build a decision tree (do not need to show any computation and do not need to be optimal).
- (2) Is this tree optimal? Explain in less than two sentences. If not, draw an optimal tree.

Problem 6: Bayesian Network



- (1) $P(+u|+e)$, using elimination.
- (2) Judging the independence of $(E, H|U)$, $(T, U|E, H)$, $(I, H|E)$, $(I, E|U)$.

Problem 7: Learning

7.1 ANN

$$(w_0, x_0, w_1, x_1, w_2, x_2) = (-3, 1, 2, -1, 4, 2)$$

$$f(\mathbf{x}) = \sigma(\mathbf{w}^T \mathbf{x})$$

Compute $\frac{\partial f}{\partial w_0}, \frac{\partial f}{\partial w_1}, \frac{\partial f}{\partial w_2}$.

7.2 CNN

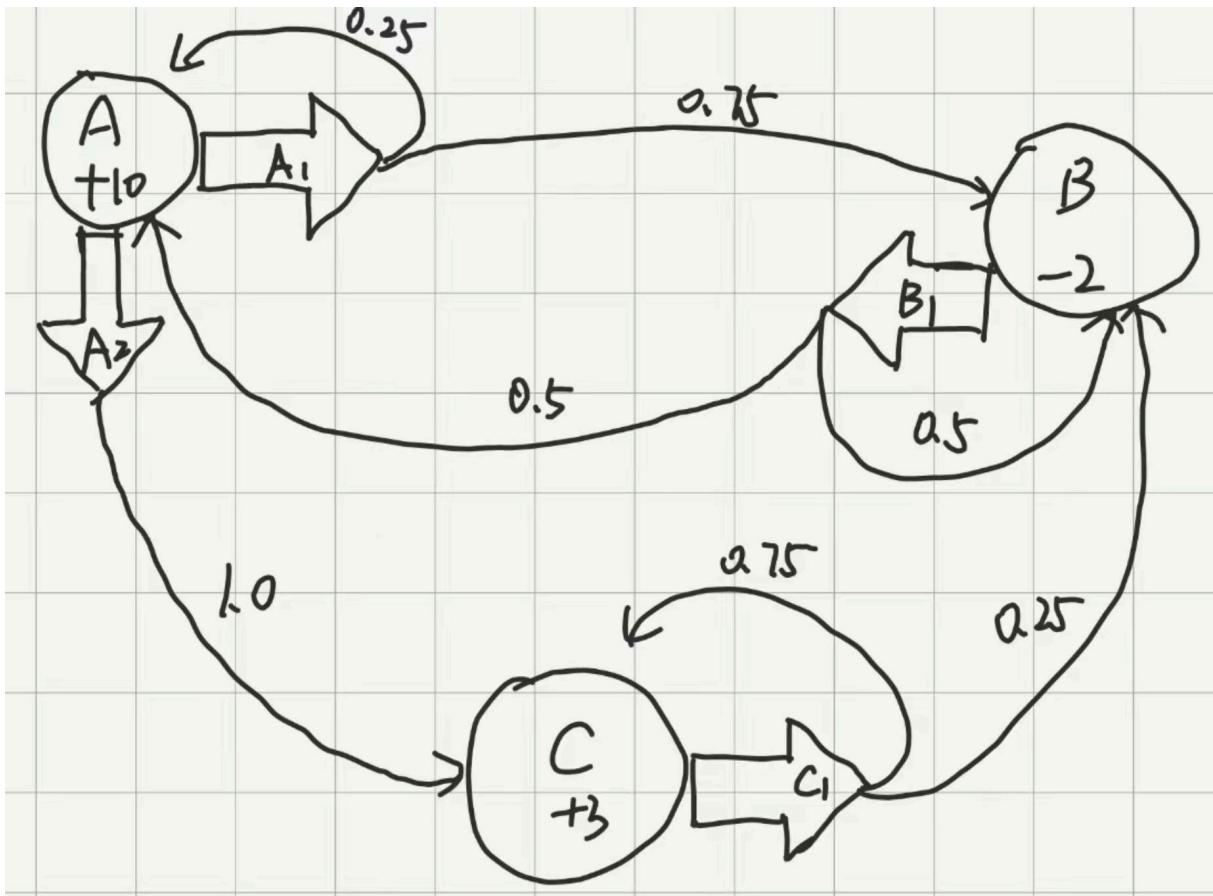
7.2.a

$$\text{Input} = \begin{pmatrix} 0 & 1 & 2 & 1 & -1 \\ 3 & 5 & 1 & 3 & 4 \\ 2 & 4 & 2 & 1 & 2 \\ 3 & 1 & 4 & 5 & 0 \\ 6 & 7 & -2 & 2 & 1 \end{pmatrix}, \text{Filter} = \begin{pmatrix} -1 & 0 & -1 \\ -2 & 1 & 1 \\ 2 & 1 & 0 \end{pmatrix}$$

stride=2, no padding, compute the output.

7.2.b

max-pooling, compute the output of Input.

Problem 8: MDP

$$U_1(s) = R(s)$$

$$U_{i+1}(s) = R(s) + \gamma \max_a \sum_{s'} T(s, a, s') U_i(s')$$

With $\gamma = 0.9$, compute:

(1) $U_2(A), U_2(B), U_2(C)$;

(2) $U_3(A), U_3(B), U_3(C)$